HELLO, I AM

Ahmad Rianul Qauliah

Geophysics

A Geophysics Graduate from Universitas Hasanuddin with a strong academic foundation in subsurface data interpretation, spatial data management, and risk mitigation analysis. Currently actively seeking professional opportunities in technical operations, geospatial mapping, the energy industry, or environmental risk management, where I can apply my analysis and field execution skills to achieve strategic goals.

About Me

Profile

A dedicated Geophysics graduate focused on subsurface exploration and natural hazard mitigation. Proficient in operating geophysical instruments such as microtremors, Ground Penetrating Radar (GPR), and seismics, with experience performing subsurface data interpretation for various academic projects.

In addition to geophysical instrument expertise, possesses field experience as a field assistant, responsible for spatial data acquisition from Total Station and Drones for mapping. Currently seeking professional opportunities in the mining, energy, and disaster mitigation industries that challenge abilities and broader experience.

3.75
GPA (IPK)

7+
Relevant Licenses & Certifications

5+
Practicum & Research Assistant

1
Published Book (Co-Author)

Education

UNIVERSITY 2020 - 2026

Universitas Hasanuddin
Bachelor of Geophysics | GPA: 3.75

- **Scholarship:** Recipient of Academic Achievement Scholarship from East Luwu Regency Government (2020–2024).
- **Book:** Co-author of Book *"Prediction of Tracks and Speed of Tropical Cyclone Vortex"* ISBN: 9789795304852 (Unhas Press, 2023).

Skills & Competencies

Technical Skills

- Subsurface Data Analysis
- Petrophysical Evaluation
- Seismic Data Processing
- Machine Learning (Python)
- GIS & Spatial Analysis
- HSE Awareness
- Topographic Surveying
- Drone Photogrammetry
- Real-time Data Monitoring
- Seismic Interpretation
- Data Acquisition
- Seismology

Software & Tools

- Python
- ArcGIS & QGIS
- Agisoft Metashape & Surfer
- Microsoft Office

Soft Skills & Languages

Work Character

- Analytical Thinking
- Adaptability
- Work Discipline
- Team Collaboration
- Time Management

Languages

Indonesian	Native
English	Passive

Experience & Academic Projects

July 2026 - Present

Summer Intern

PioPetro | Midland, Texas (Remote)

- Internship activities and details will be updated soon.

June 2026 - Present

Kaderisasi Pertambangan (Mining Preparation Mentee)

PT Geo Mining Berkah (GMB) | Jakarta Selatan

- Mentorship program details and activities will be updated soon.

Juni 2026

Interactive Web Application

SPAC-ARQ Processing Tool (Streamlit)

An interactive web application built with Python to automate and visualize the extraction of Rayleigh wave dispersion curves from passive seismic (microtremor) data using the SPAC (Spatial Autocorrelation) method. Supporting generalized array geometries (equilateral triangular and circular layouts up to 20 sessions), the app simplifies complex geophysical signal processing, offering geophysicists a robust, accessible tool for 1D shear-wave velocity (Vs) profiling and site characterization.

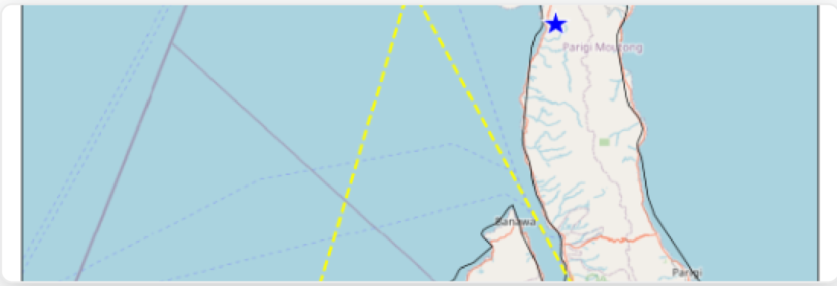
- Integrates automated time-window quality control (based on RMSE outlier detection) and a two-stage stacking algorithm to reduce data processing time from hours to minutes.
- Allows users to upload raw seismic files, filter noisy windows, and adjust frequency-cut bounds in real-time.
- Instantly exports the fitted dispersion curves through an intuitive and highly interactive visual interface.

Jun 2025 - Sep 2025

Student Project

EarthScope 2025 Seismology Skill Building Workshop

- Analyzed seismic wave characteristics and mapped the 2018 Palu earthquake using Python (ObsPy) and implemented algorithms from the localmag repository.
- Processed and filtered large regional seismicity data catalogs to focus analysis specifically on 2 main events.
- Interpreted waveform data from monitoring stations to measure earthquake source parameters, successfully estimating a minimum rupture duration of 42 seconds.
- Developed scripts to visualize epicenter data into seismicity distribution maps for simplified fault pattern and regional tectonic analysis.

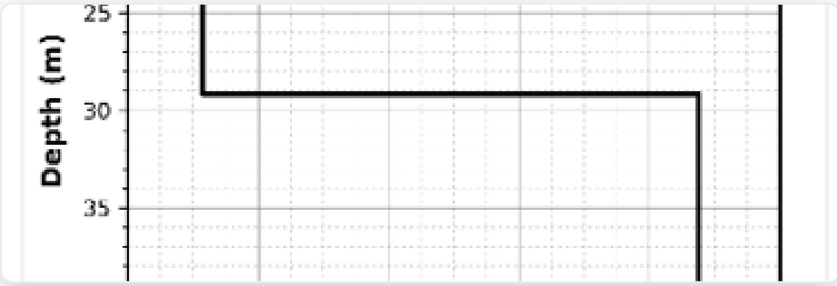


Januari 2025

Data Acquisition Operator & Field Technician

Lab Geofisika Padat, Universitas Hasanuddin

- Executed passive seismic data acquisition using SmartSolo IGU-16HR 3C 2Hz sensors on a 100-meter line with an equilateral triangular SPAC geometric array configuration (4.62 m radius) in Lonjoboko Village, Gowa Regency.
- Processed microtremor recordings by extracting dispersion curves through spatial autocorrelation analysis matched with the zeroth-order Bessel function of the first kind (J_0).
- Performed 1D subsurface modeling by applying Monte Carlo inversion algorithms on dispersion curves to estimate shear-wave velocity (Vs) profiles and determine bedrock depth.
- Characterized 5 subsurface stratigraphic layers and calculated a Vs30 value of 302.47 m/s to determine site classification as Site Class D (Medium Soil) based on SNI 1726:2019 standards.



Desember 2024

Data Acquisition Operator (GPR)

Lab Geofisika Padat, Universitas Hasanuddin

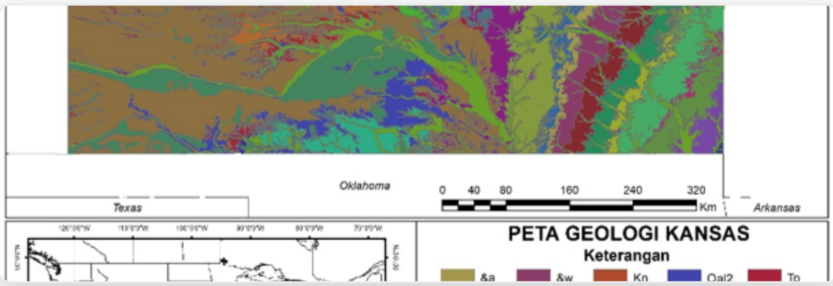
- Operated Ground Penetrating Radar (GPR) to perform non-destructive shallow subsurface mapping to identify suspected building remnants at Benteng Somba Opu.
- Executed field survey design by collecting data across 14 measurement profiles.
- Processed raw radargram data using Geoprojector software to map object distribution without conventional excavation.
- Analyzed and interpreted high-amplitude anomalies on radargram profiles to identify fort brick structures at depths of 0.30 to 3 meters with a permittivity function of 5.5–5.7.

Nov 2024 - Des 2024

Petrophysics Analysis (Case Study)

Lab Geofisika Padat, Universitas Hasanuddin

- Operated Interactive Petrophysics V3.5 software to evaluate Triple Combo Log data from the Wellington-KGS-1-32 well, successfully identifying hydrocarbon prospect zones at a depth of 3,422 feet (ft) with an average Gamma Ray value of 41.399 API.
- Conducted quantitative calculations of petrophysical parameters in target zones, yielding water saturation (Sw) of 0.839, effective porosity (PHIE) of 0.0657, and total porosity (PHIT) of 0.0842.
- Validated dominant formation lithology using NPHI/RHOB crossplot analysis, accurately confirming sandstone (0.19363%) and dolomite (0.24281%) as the main reservoir components.



Jan 2024 - Nov 2024

Field Assistant for Mapping & Drone Methods

Departemen Geofisika, Universitas Hasanuddin

- Supervised and guided over 220 Geophysics students in executing spatial data acquisition using total station, theodolite, and drones.
- Provided training to practitioners on using Agisoft Metashape for drone photogrammetry processing and ArcGIS/QGIS for data management, digitization, and mapping.
- Provided technical guidance on topographic survey methodologies, specifically the traverse method, to ensure accurate boundaries and coordinate determination.
- Demonstrated the application of Surfer software for grid-based interpolation to produce detailed contour maps and 2D surface models.
- Executed data quality assurance and control (QA/QC) on all field data and provided performance evaluations and grading.

Apr 2022 - Sep 2024

Head of Academic Division

Himpunan Ahli Geofisika Indonesia Student Chapter Universitas Hasanuddin (HAGI SC UH)

- Guest speaker at Learning Together I with HAGI SC UH members (15+ attendees) on 'Common Spatial Data in Python'.
- Responsible for designing and executing Learning Together II (Introduction to Basic Geoelectrics: Processing Geoelectrical Data with Res2Dinv) with 10+ participants.
- Produced 1-2 monthly educational contents on Geophysics for HAGI SC UH social media.

Juli 2023 - Agustus 2023

Geophysics Internship

BBMKG Wilayah IV Makassar (BMKG)

- Monitored real-time regional seismic activity using SeisComP to quickly identify structural anomalies.
- Managed the daily monitoring database of 50+ geophysical stations by recording, verifying, and cataloging earthquake events to maintain data integrity.
- Processed and interpreted initial seismic data by manually picking P and S-wave phases to determine hypocenter, epicenter, depth, and magnitude parameters.

Nov 2021 - Agu 2023

Academic Division Member

Himpunan Mahasiswa Geofisika Universitas Hasanuddin (HMGF UH)

- Executed several division programs such as Geophysical Instrumentation Studies and Geoelectrical Surveys.
- Successfully executed Geophysical Instrumental Studies (GIS) on Automatic Weather Stations with 30+ participants.

Desember 2022 - Februari 2023

KKN Tematik Kebencanaan Universitas Hasanuddin (Community Service)

Desa Limbong Wara, Luwu Utara

- **Village Mapping (PAPEDA):** Compiled administrative and infrastructural maps of Limbong Wara Village based on specific spatial data using GIS software for accurate village spatial planning.
- **Disaster Risk Mapping:** Analyzed local topography to generate disaster risk zoning maps to support preparedness education for residents.

Nov 2022 - Des 2022

Bawakaraeng Slopes Mapping Project

Lab Geoinformatika, Universitas Hasanuddin

- Processed raw photogrammetry data from drone mapping of the slopes of Mount Bawakaraeng to generate accurate 3D topographic models of the terrain.
- Performed spatial data reconstruction using the WGS 84 coordinate system and operated Agisoft Metashape software to produce Dense Clouds, Orthophotos, and Orthomosaic maps.
- Generated and analyzed Digital Elevation Models (DEM) and Digital Terrain Models (DTM) to represent terrain relief.

Licenses, Certifications & Publications

PUBLISHED BOOK

Prediction of Tracks and Speed of Tropical Cyclone Vortex

ISBN: 9789795304852 | Penerbit: Unhas Press (2023)

Collaborative book (as co-author) discussing modeling and predicting tracks and speed of tropical cyclones using meteorological and geophysical parameters.

Well Operations Crew Resource Management (WOCRM)
International Well Control Forum (IWCF)
2026

Level 1 IWCF Programme
International Well Control Forum (IWCF)
2026

Safety Compliance: CSMS Breakdown
Sahara.id
2026

Pre-POP Training (First Line Operational Supervisor)
PT Grow Mining Corp Indonesia
2026

Seismology Skill Building Workshop
EarthScope Consortium
2025

Oil & Gas Industry Operations and Markets
Duke University / Coursera
2025

Get Mapping Quickly with QGIS
Map Academy / Udemy
2024

Workshop Trio Migas.id
Reservoir, Drilling & Production Techniques
2024

Seismic Data Processing
Open Learning USM MOOC
2024

Petroleum School Migas.id Batch 8
Migas.id
2023

Programming for Everybody (Python)
University of Michigan / Coursera
2022

Python for Data Science & GIS Analysis
Geosoftware Community
2021

National Model Making Finalist - Transfest
(HIMASEATRANS) FTK - ITS Surabaya
2019

3rd Place - National Science Olympiad (OSN) in Earth Science, East Luwu
Olimpiade Sains Nasional
2018

Contact Me

Let's Connect!

I am always open to discussing internship/employment opportunities, collaborative projects, or simply exchanging thoughts on geophysics and the energy industry.

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